

Dr.Doc Musculo:

Welcome to Musculoskeletal Pathology

Skin Pathology

Skin diseases are common and diverse. The range runs from simple inflammatory disorders (rashes, urticaria, eczema) all the way to highly malignant neoplasms (melanoma).

Two ways the skin gets involved

☐ Intrinsic to the skin: the disease starts and stays in the skin (psoriasis, warts, basal cell carcinoma).

☐ A window onto a body-wide disease: the skin shows a manifestation of something that involves many tissues.

☐ Why this matters

When you see a skin lesion, ask two questions: "Is this a skin-only problem?" and "Could this be the visible tip of a systemic disease?" Examiners love that second question.

In skin pathology, the clinical history, the gross appearance, and the distribution of lesions are often as important as the microscopic findings in reaching a specific diagnosis.

Why history, gross look, and distribution matter

☐ The reasoning behind it

The skin has a limited number of ways to react. Many different diseases produce a similar-looking pattern under the microscope. The slide alone may say "spongiotic dermatitis", but only the story and the location tell you which kind. Think of histology as confirming a pattern, and the clinical picture as choosing the diagnosis within it.

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Dr. Doc Tutorial

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📖 Acantholysis

📖 Definition: Loss of the intercellular connections between keratinocytes, so the cells lose cohesion.

📖 What you see?

Rounded, detached (acantholytic) keratinocytes floating in a blister cavity.

📖 Pemphigus vulgaris. The layer just above the basal cells separates, giving a suprabasal acantholytic blister.

📖 Pemphigus foliaceus. Separation at the stratum granulosum: a superficial, subcorneal blister.

❓ Cause. IgG autoantibodies against desmosomal proteins

(desmoglein 1 and 3), a type II hypersensitivity.

❓ Bullous pemphigoid. The blister is subepidermal, and the roof lacks acantholysis.

❓ Spongiosis

Definition: Intercellular edema of the epidermis.

❓ What you see?

Fluid pushes keratinocytes apart and stretches the intercellular bridges, so they look more prominent and the epidermis looks "spongy". The cells stay attached to one another.

❓ Typical of acute eczematous dermatitis: red, papulovesicular, oozing, crusted lesions.

❓ Accompanied by a superficial perivascular lymphocytic infiltrate, papillary dermal edema, and mast cell degranulation. Eosinophils may be present.

❓ With continued fluid build-up, it can progress to frank blister formation.

❓ Papillomatosis

Definition: hyperplasia of the papillary dermis, with elongation or widening of the dermal papillae, producing surface elevation.

❑ Classic in verrucae (warts), caused by HPV. Papillary epidermal proliferation often radiates symmetrically "like the points of a crown".

❑ Warts also show koilocytosis (halos around infected nuclei).

❑ Grossly, common warts have a rough, pebble-like surface.

❑ Lentiginous

Definition: a linear pattern of melanocyte proliferation along the epidermal basal cell layer. It can be a reactive change or part of a melanocyte neoplasm.

❑ Lentigo: benign, linear (non-nested) melanocytic hyperplasia just above the basement membrane; small tan-brown macules.

❑ Dysplastic nevus: single nevus cells begin to replace the normal basal layer, called lentiginous hyperplasia, seen as the flat "shoulder" around the central raised part.

❑ Contrast, freckles: melanocyte number is normal; only the melanin in basal keratinocytes increases.

❑ Mix-up busters

❑ Spongiosis vs acantholysis: in spongiosis cells are pushed apart by fluid but still connected. In acantholysis the connections are gone.

❑ Acanthosis vs papillomatosis: acanthosis thickens the epidermis. Papillomatosis enlarges the dermal papillae.

❑ Hyperkeratosis vs parakeratosis: hyperkeratosis is thickness of the corneum. Parakeratosis is nuclei retained in it.

★High-Yield Points

1. History, gross appearance, and distribution are as important as histology. ❑
2. The 5 mm rule. Macule/patch, papule/nodule, and vesicle/bulla all split at 5 mm. ❑
3. A wheal is transient and caused by dermal edema. ❑
4. Excoriation is a scratch. Lichenification is thickened skin from rubbing. ❑
5. Acanthosis is epidermal hyperplasia. Hyperkeratosis is a thick corneum. ❑
6. Parakeratosis means nuclei in the corneum, and it is normal on mucous membranes. ❑
7. Spongiosis is intercellular edema, the hallmark of eczema. ❑
8. Acantholysis is loss of cell-to-cell adhesion, seen in pemphigus and not in pemphigoid. ❑
9. Papillomatosis (warts) affects dermal papillae. Lentiginous means linear basal melanocytes (lentigo, dysplastic nevus). ❑